Advanced Econometrics

Censored data models

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Lecture Overview

- Tobit (Censored) Model
- Truncated Regression Model
- Heckman Sample Selection Model (brief)
- Key goal: **Understand differences and when to apply each model**

Limited Dependent Variables (LDV)

- Dependent variable restricted to an interval
- Examples:
  - $[0, \infty)$ (e.g. expenditures)
  - $[a, b]$ (bounded outcomes)
- Standard OLS may be inappropriate

Three Key Models: Very Simple Difference

- **Tobit model** = we see **all people**, but for some of them the outcome is **stuck at a limit**
  - Example: spending on shoes
  - Many people spend \$0
  - \$0 is a real choice
  - We observe everyone
- **Truncated regression** = we see **only part of the sample**
  - Example: test scores only for students with score ≥ 40
  - Students below 40 are not in the dataset at all
  - So part of the population is missing
- **Heckman model** = we see everyone, but the outcome is observed only for a **selected group**, and that selection is **not random**
  - Example: wages of married women
  - For women not working, wage is not really 0
  - We simply do not observe the wage they would earn

In stupid-student language:

- **Tobit:** everyone is here, but many values hit the wall
- **Truncated:** some people are missing completely
- **Heckman:** outcome is missing for a non-random group

Tobit Model (Censored Regression)

- Used when:
  - Dependent variable is **limited (censored)** at some value (often 0)
  - Many observations are exactly at that limit
- Key idea:
  - There is a **hidden (latent) variable** y_i^* we would like to observe
  - But we only observe:
    - 0 if the value is too low
    - the true value if it is above the limit
- Example:
  - Number of affairs in past year
  - Many people report **0** (no affairs)
  - But the decision is real: people **choose zero**
- Intuition (very simple):
  - Everyone is observed
  - But many values are **piled up at zero**

Model:

$$y_i^* = X_i\beta + \varepsilon_i$$

$$y_i = \begin{cases} 0 & \text{if } y_i^* \leq 0 \\ y_i^* & \text{if } y_i^* > 0 \end{cases}$$

Why Not OLS? (Very Important)

- OLS assumes:
  - The dependent variable is **fully observed**
  - The relationship is **linear for all values**
- Problem in Tobit setting:
  - Many observations are **censored at 0**
  - OLS treats all zeros as normal values
  - But zeros are **special (boundary outcomes)**
- What goes wrong?
  - **Bias:** coefficients are systematically wrong
  - **Inconsistency:** even with more data, estimates do not converge to true values
- Intuition (simple):
  - OLS tries to fit a straight line through many zeros
  - This **pulls the regression line down**
  - Result: underestimation of true effects
- When is it really bad?
  - When many observations are at the boundary (e.g. 70% zeros)
- Exception:
  - If almost no observations are censored $\rightarrow$ OLS $\approx$ Tobit

Corner Solution vs Censoring (Key Concept)

Corner Solution:

- Zero is a **real, optimal decision**
- Individuals **choose** to be at the boundary
- Example:
  - Spending on charity = 0
  - Buying no shoes
- Interpretation:
  - 0 means exactly 0
  - No hidden information

Censoring:

- Observed value does **not equal true value**
- Data is **cut off or grouped**
- Example:
  - Income reported as "4k+"
  - Survey caps answers at a threshold
- Interpretation:
  - Observed value = **minimum or maximum**
  - True value could be higher (or lower)
- Corner solution = **choice**
- Censoring = **data problem**

obit Marginal Effects

- Unlike OLS:
  - In OLS $\rightarrow$ one marginal effect
  - In Tobit $\rightarrow$ **multiple marginal effects**
- Why?
  - Because we model:
    - **whether** $y > 0$
    - **how much** y is when $y > 0$

Tobit Marginal Effects (Very Important)

Three key marginal effects (corner solution case):

1. **Unconditional expectation:**
$$\frac{\partial E[y]}{\partial x}$$
   - Effect of x on overall expected value of y
   - Includes both zeros and positive values
2. **Conditional expectation:**
$$\frac{\partial E[y|y > 0]}{\partial x}$$
   - Effect of x only for individuals with $y > 0$
   - "Among those who participate"
3. **Probability of being uncensored:**
$$\frac{\partial P(y > 0)}{\partial x}$$
   - Effect of x on probability of having positive outcome
   - Usually interpreted in percentage points

Tobit Marginal Effects (Very Important)

Key idea:

- Tobit separates:
  - participation (zero vs non-zero)
  - intensity (how large the value is)

Interpretation: Simple Intuition

- One variable (e.g. age) affects TWO things:
  - **Participation:**
    - Does age affect whether someone has affairs at all?
  - **Intensity:**
    - If they do have affairs, does age affect how many?
- That is why we have multiple marginal effects
- **Very simple takeaway:**
  - Tobit = **who participates** + **how much**

Truncated Regression (Missing Data)

- Used when:
  - Part of the population is **completely missing from the dataset**
  - We observe only a **subset of individuals**

- Key idea:
  - Some observations are **not recorded at all**
  - They are not zero — they are **invisible**

  Example:
  - Students in a special program
  - Only students with score ≥ 40 are included
  - Students with score < 40 are **not in the data**

  What is the problem?
  - We do not see the full distribution
  - Data is **cut from one side** (left or right)

  Important difference from Tobit:
  - Tobit: we see zeros (they exist in data)
  - Truncated: we do **not see missing observations at all**

  Very simple intuition:
  - Tobit $=$ data hits a wall
  - Truncated $=$ some people are missing completely

Truncation vs Censoring (Very Important)

Truncation:

- Some observations are **completely missing**
- Data is **cut off** below or above a threshold
- We do not even know these observations exist
- Dataset of incomes but only for people earning > 2000
- Only billionaires in the dataset (wealth ≥ 1 billion)
- Students admitted only if score ≥ 40

Censoring:

- All observations are present
- But some values are **reported as limits**
- True value is partially hidden
- Income reported as "≥ 4000"
- Survey: "0 or more than 10"
- Number of affairs: many observed as 0

Key difference:

- Truncation = **missing people**
- Censoring = **hidden values**

Self-Selection (Very Simple Idea)

- Self-selection means:
  - People **choose** whether to be in the sample
- Problem:
  - The group we observe is **not random**
  - It is **systematically different**

Example:

  - Wages of women
  - We only observe wages for women who **work**
  - Women who do not work are missing wages

Why is this a problem?

  - Working women may be:
    - more educated
    - more motivated
  - So the sample is **biased**

Very simple intuition:

  - Data is not wrong
  - But the sample is **not representative**

Self-Selection (More Examples)

- Self-selection = people **choose** whether to be observed
  Classic example:
  - Wages of women
  - We only observe wages for women who **work**

  More examples:
  - **Gym attendance:**
    - We observe time spent at the gym
    - But only for people who **go to the gym**
  - **Online course performance:**
    - We observe grades only for students who **finish the course**
    - Weak students may drop out earlier
  - **Loan approval:**
    - We observe repayment only for people who **got a loan**
    - High-risk applicants may be rejected
  - **Survey data:**
    - Only people who **choose to respond** are observed
    - Responses may be biased
  - **Job applications:**
    - We observe wages only for people who **apply and get jobs**
    - Others are missing

  Key problem:
  - Observed group is **not random**

Heckman Sample Selection Model

- Used when:
  - Selection into the sample is **non-random**
  - We observe outcomes only for a **selected group**
- Key idea:
  - Two separate processes:
    - **1. Selection** (who is observed)
    - **2. Outcome** (value of dependent variable)
- Two equations:
  - **Selection equation:**
    - Models probability of being in the sample
    - Example: working vs not working
  - **Outcome equation:**
    - Models the dependent variable
    - Example: wage level

Example:

- Wages of married women
- We only observe wages for women who work
- Non-working women have **missing (not zero) wages**

Very simple intuition:

- First: who shows up?
- Second: what value do they have?

Heckman Model Example (Wages of Women)

- We want to study: What determines wages of married women?
- Problem:
  - We observe wages only for women who **work**
  - Many women do **not participate** in the labor market
- Important:
  - Observed wage $= 0$ does **NOT** mean true wage is 0
  - It means: wage is **unobserved (missing)**

Why is this a problem?

- Women who work are **not random**
- They may differ in:
  - education
  - experience
  - motivation

Selection bias:

- We estimate wages only for a **selected group**
- Results do not represent all women

Example intuition:

- If only highly educated women work
- Estimated wages will be **too high**

Use Heckman model to correct for selection

Heckman Model Structure (Two-Step Idea)

Step 1: Selection equation (Probit) First model: **who is observed**

$$D^* = Z\gamma + u$$

- Models **whether we observe the outcome**
- $D = 1$ if individual is in the sample (e.g. works)
- $D = 0$ if not observed (e.g. does not work)
- Z = variables affecting participation (e.g. age, children, income)

Step 2: Outcome equation Second model: **what value they have**

$$y = X\beta + \varepsilon$$

- Models the **actual outcome**
- Example: wage level
- Observed only when $D = 1$
- X = variables affecting the outcome (e.g. education, experience)

Key problem:

- Errors u and ε may be **correlated**
- This creates **selection bias**

Model Choice Summary

Model	Problem	Example
Tobit	Censoring, many zeros	Charity spending
Tobit	Boundary values	Affairs (Fair data)
Truncated	Missing observations	Income $>$ 2000 only
Truncated	Sample cut off	Test score $\geq$ 40
Heckman	Self-selection	Wages (only workers)
Heckman	Non-random sample	Loan approval

Memory: Tobit $=$ wall   Truncated $=$ missing   Heckman $=$ selection

Decision Guide

1. Is dependent variable limited?
2. Are observations missing (truncation)?
3. Is there selection bias?
4. Is zero a true outcome or hidden value?

Key Takeaways

- Choosing the correct model is crucial
- Tobit $\neq$ Truncated $\neq$ Heckman
- Interpretation differs across models
- Understanding data structure is essential

Thank you